

CBSE Class 12 Chemistry — Solutions

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQs / Assertion-Reason / Very Short Answer)

- Q1.** Which of the following is an example of a solid-in-solid solution? (a) Sea water (b) Sugar solution (c) Smoke (d) 22 carat gold [PYQ 2021] [Confidence: Very High]
- Q2.** The solubility of gases in liquids decreases with a rise in temperature because dissolution is an: (a) endothermic and reversible process (b) exothermic and reversible process (c) endothermic and irreversible process (d) exothermic and irreversible process [PYQ 2021] [Confidence: Very High]
- Q3.** On mixing 20 mL of acetone with 30 mL of chloroform, the total volume of the solution is: (a) < 50 mL (b) = 50 mL (c) > 50 mL (d) = 10 mL [PYQ 2021] [Confidence: High]
- Q4.** An azeotropic solution of two liquids has a boiling point lower than either of the two when it: (a) shows a positive deviation from Raoult's law. (b) shows a negative deviation from Raoult's law. (c) shows no deviation from Raoult's law. (d) is saturated. [PYQ 2021] [Confidence: Very High]
- Q5.** Which one of the following pairs will form an ideal solution? (a) Chloroform and acetone (b) Ethanol and acetone (c) *n*-hexane and *n*-heptane (d) Phenol and aniline [PYQ 2021] [Confidence: Very High]
- Q6.** An unknown gas **X** is dissolved in water at 2.5 bar pressure and has a mole fraction of 0.04 in solution. The mole fraction of **X** gas when the pressure of the gas is doubled at the same temperature is: (a) 0.08 (b) 0.04 (c) 0.02 (d) 0.92 [PYQ 2021] [Confidence: High]
- Q7.** Which of the following formulas represents Raoult's law for a solution containing a non-volatile solute? (a) $P_{\text{solute}} = P_{\text{solute}}^{\circ} \cdot X_{\text{solute}}$ (b) $P = K_H \cdot X$ (c) $P_{\text{Total}} = P_{\text{solvent}}^{\circ} \cdot X_{\text{solvent}}$ (d) $P_{\text{solute}} = P_{\text{solvent}} \cdot X_{\text{solvent}}$ [PYQ 2021] [Confidence: Very High]
- Q8.** A 0.5 M aqueous solution of glucose is isotonic with: (a) 0.5 M **KCl** solution (b) 0.5 M **CaCl₂** solution (c) 0.5 M urea solution (d) 1 M solution of sucrose [PYQ 2021] [Confidence: High]
- Q9.** Which pair of solutions is isotonic at the same temperature? (a) 0.1 M urea and 0.1 M **NaCl** (b) 0.1 M **NaCl** and 0.1 M **Na₂SO₄** (c) 0.1 M urea and 0.2 M **MgCl₂** (d) 0.1 M **Na₂SO₄** and 0.1 M **Ca(NO₃)₂** [PYQ 2021] [Confidence: Very High]
- Q10.** The medical condition of water retention or puffiness due to high salt intake occurs due to: (a) Reverse osmosis (b) Osmosis (c) Diffusion (d) Dialysis [PYQ 2021] [Confidence: Medium]
- Q11.** Assertion (A): Aquatic species are more comfortable in cold water than in warm water. Reason (R): The solubility of oxygen in water increases with a decrease in temperature. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ 2018] [Confidence: Very High]

Q12. Assertion (A): Elevation in boiling point is a colligative property. Reason (R): The elevation in boiling point depends on the number of moles of solute dissolved in the solvent, not on its nature. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ 2016] [Confidence: Very High]

Q13. Assertion (A): A mixture of phenol and aniline shows a negative deviation from Raoult's law. Reason (R): Intermolecular hydrogen bonding between phenol and aniline is stronger than the respective intermolecular hydrogen bonding in pure phenol and pure aniline. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ 2021] [Confidence: High]

Q14. Define Molality (m). [PYQ 2017] [Confidence: Very High]

Q15. What is the Van't Hoff factor (i) for a dilute aqueous solution of completely dissociated $MgCl_2$? [PYQ 2016] [Confidence: Very High]

Q16. Define abnormal molar mass. [PYQ 2020] [Confidence: High]

Q17. Define osmotic pressure (Π). [PYQ 2017] [Confidence: Very High]

Q18. What are azeotropes? [PYQ 2014] [Confidence: Very High]

Q19. What is the effect of temperature on the solubility of a solid solute (like glucose) in water if the dissolution process is endothermic? [PYQ 2023] [Confidence: Medium]

Q20. Out of 1 M glucose and 1 M $NaCl$, which will have a higher boiling point and why? [Practice] [Confidence: Very High]

2-Mark Questions (Short Answer)

Q21. Give reasons for the following: (i) People living at higher altitudes often suffer from a condition called anoxia. (ii) Water stops boiling when sugar is added to boiling water. [PYQ 2019] [Confidence: Very High]

Q22. State Raoult's law for a solution containing volatile components. What is the fundamental similarity between Raoult's law and Henry's law? [PYQ 2020] [Confidence: Very High]

Q23. Write two distinct differences between ideal solutions and non-ideal solutions. [PYQ 2019] [Confidence: High]

Q24. What happens when acetone is added to pure ethanol? Explain in terms of intermolecular forces and deviation from Raoult's law. [PYQ 2020] [Confidence: High]

Q25. Calculate the mass of ascorbic acid ($C_6H_8O_6$, Molar mass = 176 g mol^{-1}) to be dissolved in 75 g of acetic acid to lower its freezing point by 1.5°C . ($K_f = 3.9 \text{ K kg mol}^{-1}$) [PYQ 2020] [Confidence: Very High]

Q26. For a 5% (by mass/volume) solution of urea (Molar mass = 60 g mol^{-1}), calculate the osmotic pressure at 300 K. ($R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$) [PYQ 2020] [Confidence: High]

Q27. Identify which liquid will have a higher vapour pressure at 90°C if the boiling points of two pure liquids A and B are 140°C and 180°C , respectively. Justify your answer. [PYQ 2020] [Confidence: Medium]

Q28. What happens when a pressure greater than the osmotic pressure is applied on the solution side separated from a pure solvent by a semipermeable membrane? State one practical application of this phenomenon. [PYQ 2020] [Confidence: Very High]

Q29. Define Ebullioscopic constant (K_b) and Cryoscopic constant (K_f). Write their SI units. [PYQ 2014] [Confidence: High]

Q30. Human blood cells are isotonic with a 0.9% (m/v) sodium chloride solution. Predict what happens if we place blood cells in: (a) 1.2% sodium chloride solution (b) 0.4% sodium chloride solution [PYQ 2016] [Confidence: Very High]

Q31. 3.9 g of benzoic acid dissolved in 49 g of benzene shows a depression in freezing point of 1.62 K. Calculate the Van't Hoff factor (i) and predict the nature of the solute (associated or dissociated). (Given: Molar mass of benzoic acid = 122 g mol^{-1} , K_f for benzene = 4.9 K kg mol^{-1}) [PYQ 2015] [Confidence: High]

Q32. Why are the tanks used by scuba divers filled with air diluted with helium (11.7% helium, 56.2% nitrogen, and 32.1% oxygen)? [PYQ 2022] [Confidence: Very High]

3-Mark Questions (Short Answer)

Q33. State Henry's law. Calculate the solubility of CO_2 in water at 298 K under 760 mm Hg . (K_H for CO_2 in water at 298 K is $1.25 \times 10^6\text{ mm Hg}$). [PYQ 2020] [Confidence: Very High]

Q34. The vapour pressure of pure water at a certain temperature is 23.80 mm Hg . If 1 mole of a non-volatile non-electrolytic solute is dissolved in 100 g of water, calculate the resultant vapour pressure of the solution. [PYQ 2023] [Confidence: Very High]

Q35. The vapour pressures of pure liquid X and pure liquid Y at 25°C are 120 mm Hg and 160 mm Hg respectively. If equal moles of X and Y are mixed to form an ideal solution, calculate the total vapour pressure of the solution. [PYQ 2023] [Confidence: High]

Q36. Calculate the boiling point of a solution when 4 g of MgSO_4 (Molar mass = 120 g mol^{-1}) was dissolved in 100 g of water, assuming MgSO_4 undergoes complete ionisation. (K_b for water = $0.52\text{ K kg mol}^{-1}$, Boiling point of pure water = 373.15 K). [PYQ 2016] [Confidence: Very High]

Q37. When 19.5 g of $\text{F}-\text{CH}_2-\text{COOH}$ (Molar mass = 78 g mol^{-1}) is dissolved in 500 g of water, the depression in freezing point is observed to be 1°C . Calculate the degree of dissociation (α) of the acid. (Given: K_f for water = $1.86\text{ K kg mol}^{-1}$) [PYQ 2023] [Confidence: High]

Q38. A solution containing 15 g of urea (Molar mass = 60 g mol^{-1}) per litre of solution in water has the same osmotic pressure (isotonic) as a solution of glucose (Molar mass = 180 g mol^{-1}) in water. Calculate the mass of glucose present in one litre of its solution. [PYQ 2014] [Confidence: High]

Q39. A 10% solution (by mass) of sucrose in water has a freezing point of 269.15 K. Calculate the freezing point of 10% glucose in water if the freezing point of pure water is 273.15 K. (Given: Molar mass of sucrose = 342 g mol^{-1} , Molar mass of glucose = 180 g mol^{-1}) [PYQ 2017] [Confidence: Very High]

Q40. When 2.56 g of sulphur was dissolved in 100 g of CS_2 , the freezing point lowered by 0.383 K. Calculate the formula of sulphur (S_x). (K_f for $\text{CS}_2 = 3.83 \text{ K kg mol}^{-1}$, Atomic mass of sulphur = 32 g mol^{-1}) [PYQ 2016] [Confidence: High]

Q41. Calculate the mass of NaCl (Molar mass = 58.5 g mol^{-1}) to be dissolved in 37.2 g of water to lower the freezing point by 2°C assuming that NaCl undergoes complete dissociation. (K_f for water = $1.86 \text{ K kg mol}^{-1}$) [PYQ 2015] [Confidence: Very High]

Q42. Define an azeotropic mixture. What type of azeotropic mixture will be formed by a solution of acetone and chloroform? Justify on the basis of the strength of intermolecular interactions that develop in the solution. [PYQ 2019] [Confidence: High]

5-Mark Questions (Long Answer / Case-Study)

Q43. (A) Define Colligative properties. Why is molality preferred over molarity for calculating colligative properties involving temperature changes? (B) Derive the relationship between the relative lowering of vapour pressure and the molar mass of a non-volatile solute. [PYQ 2015 / Practice] [Confidence: High]

Q44. (A) A 5% (by mass/volume) solution of $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ (Molar mass = 322 g mol^{-1}) is isotonic with a 2% (by mass/volume) solution of a non-electrolytic, non-volatile substance X. Find out the molecular weight of X. (Assume complete dissociation of the salt). (B) Explain with the help of a vapour pressure-temperature graph why the boiling point of a solvent elevates upon the addition of a non-volatile solute. [PYQ 2023 / Practice] [Confidence: Very High]

Q45. Case Study: Osmosis and Osmotic Pressure When a semipermeable membrane separates a solution from a pure solvent, solvent molecules spontaneously pass through the membrane into the solution. This is called osmosis. The external pressure that must be applied to the solution side to stop this flow is called osmotic pressure (Π). (A) What is reverse osmosis? State its main practical application. (1 Mark) (B) Write the mathematical relationship between osmotic pressure (Π) and the molarity (C) of the solution. (1 Mark) (C) Calculate the molar mass of a polymer if 1.0 g of the polymer dissolved in 200 mL of water produces an osmotic pressure of $2.0 \times 10^{-3} \text{ atm}$ at 300 K. ($R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$) (3 Marks) [PYQ 2021 / Practice] [Confidence: Very High]

Q46. (A) What type of deviation from Raoult's law is shown by a mixture of ethanol and acetone? Give a reason. (B) Calculate the Van't Hoff factor (i) for an aqueous solution of $\text{K}_3[\text{Fe}(\text{CN})_6]$ if its degree of dissociation (α) is 0.852. (C) Using the calculated i from part (B), what will be the boiling point of this solution if its concentration is 1 molal? (K_b for water = $0.52 \text{ K kg mol}^{-1}$) [PYQ 2023 / 2014] [Confidence: High]

Q47. Case Study: Depression in Freezing Point The addition of a non-volatile solute lowers the vapour pressure of a solvent, which consequently depresses its freezing point. (A) Write the

formula relating depression in freezing point (ΔT_f) to molality (m) and Van't Hoff factor (i). (1 Mark) (B) Out of 1 M glucose and 1 M $MgCl_2$, which solution will have a higher boiling point and why? (1 Mark) (C) The freezing point of a solution containing 5 g of benzoic acid (Molar mass = 122 g mol^{-1}) in 35 g of benzene is depressed by 2.94 K. What is the percentage association of benzoic acid if it forms a dimer in solution? (K_f for benzene = $4.9 \text{ K kg mol}^{-1}$) (3 Marks) [PYQ 2020 / Practice] [Confidence: Very High]

Q48. (A) State Henry's Law. Mention any two critical real-life applications of Henry's law. (B) Assume that Argon gas exerts a partial pressure of 6 bar. Calculate the solubility (mole fraction) of Argon gas in water. (Given: Henry's Law constant, K_H , for Argon dissolved in water is **40 kbar**). [PYQ 2022] [Confidence: High]

Q49. (A) Define an ideal solution. What are the conditions of ΔH_{mix} and ΔV_{mix} for an ideal solution? (B) The vapour pressure of pure liquids A and B are 450 mm Hg and 700 mm Hg respectively at 350 K. Find out the composition of the liquid mixture (mole fractions of A and B) if the total vapour pressure is 600 mm Hg. Also, find the composition of the vapour phase. [Practice] [Confidence: High]

Q50. (A) What is meant by negative deviations from Raoult's law? Give an example. What are the signs of ΔH_{mix} and ΔV_{mix} for negative deviation? (B) Calculate the boiling point elevation (ΔT_b) when 10 g of $CaCl_2$ (Molar mass = 111 g mol^{-1}) is added to 200 g of water. Assume $CaCl_2$ undergoes complete dissociation. (K_b for water = $0.512 \text{ K kg mol}^{-1}$) [PYQ 2017 / Practice] [Confidence: Very High]

Answer Key

Q1. (d) 22 carat gold (alloy of solid in solid). **Q2.** (b) exothermic and reversible process. Dissolution of gas releases heat. **Q3.** (a) < 50 mL. Acetone and chloroform show negative deviation (stronger H-bonding), so $\Delta V_{mix} < 0$. **Q4.** (a) shows a positive deviation from Raoult's law (forces are weaker, escapes easily). **Q5.** (c) *n*-hexane and *n*-heptane (similar polarity and structure). **Q6.** (a) 0.08. Since $P = K_H X$, doubling pressure doubles the mole fraction. **Q7.** (c) $P_{Total} = P_{solvent}^\circ \cdot X_{solvent}$. **Q8.** (c) 0.5 M urea solution. Both are non-electrolytes ($i = 1$), so same concentration gives same osmotic pressure. **Q9.** (d) 0.1 M Na_2SO_4 and 0.1 M $Ca(NO_3)_2$. Both yield 3 ions ($i = 3$), making them isotonic. **Q10.** (b) Osmosis. Fluid retention in tissues due to osmosis. **Q11.** (a) Both A and R are true and R is the correct explanation of A. **Q12.** (a) Both A and R are true and R is the correct explanation of A. **Q13.** (a) Both A and R are true and R is the correct explanation of A. **Q14.** Molality is the number of moles of solute dissolved per kilogram (1000 g) of the solvent. **Q15.** 3. ($MgCl_2 \rightarrow Mg^{2+} + 2Cl^-$) **Q16.** When the experimentally determined molar mass of a solute is higher or lower than the expected theoretical value due to association or dissociation in solution, it is termed abnormal molar mass. **Q17.** Osmotic pressure is the excess pressure applied to the solution side to just stop the flow of solvent molecules through a semipermeable membrane. **Q18.** Azeotropes are binary mixtures having the same composition in the liquid and vapour phases and boiling at a constant temperature. **Q19.** According to Le Chatelier's principle, if dissolution is endothermic ($\Delta H > 0$), solubility increases with an increase in temperature. **Q20.** 1 M $NaCl$. It dissociates into 2 ions ($i = 2$), whereas

glucose is a non-electrolyte ($i = 1$). Higher particle count leads to greater boiling point elevation.

Q21. (i) At high altitudes, partial pressure of O_2 is lower than at ground level, leading to low oxygen concentration in blood/tissues, impairing thinking (anoxia). (ii) Sugar (non-volatile solute) lowers the vapour pressure of water, thereby elevating its boiling point. Thus, it momentarily stops boiling until the new higher BP is reached. **Q22.** Raoult's law states that for a solution of volatile liquids, the partial vapour pressure of a component is directly proportional to its mole fraction. Similarity: Both state that partial vapour pressure of the volatile component (or gas) is directly proportional to its mole fraction ($P = K \cdot \chi$). **Q23.** (1) Ideal solutions obey Raoult's law; non-ideal do not. (2) For ideal, $\Delta H_{mix} = 0$ and $\Delta V_{mix} = 0$; for non-ideal, they are not zero.

Q24. Shows positive deviation. Ethanol molecules have strong intermolecular H-bonding.

Acetone molecules break these H-bonds, decreasing overall intermolecular forces, increasing vapour pressure. **Q25.** $\Delta T_f = K_f \cdot \frac{W_B \times 1000}{M_B \times W_A} \Rightarrow 1.5 = \frac{3.9 \times W_B \times 1000}{176 \times 75} \Rightarrow W_B = 5.07 \text{ g}$. **Q26.** 5% m/v = 5g in 100 mL = 50g in 1 L. $Molarity(C) = \frac{50}{60} \text{ M}$.

$\Pi = CRT = \frac{50}{60} \times 0.0821 \times 300 = 20.525 \text{ atm}$. **Q27.** Liquid A (140°C). Lower boiling point implies weaker intermolecular forces and a higher tendency to escape into the vapour phase, thus higher vapour pressure at a given temperature. **Q28.** Reverse osmosis occurs; the solvent flows from the solution back into the pure solvent side. Application: Desalination of seawater.

Q29. K_b : Elevation in boiling point when 1 mole of solute is dissolved in 1 kg of solvent. K_f : Depression in freezing point for 1 molal solution. Unit for both: K kg mol^{-1} . **Q30.** (a) In 1.2% NaCl (hypertonic), water moves out, cells shrink (plasmolysis). (b) In 0.4% NaCl (hypotonic), water moves in, cells swell and may burst. **Q31.**

$\Delta T_f = i \cdot K_f \cdot \frac{W_B \times 1000}{M_B \times W_A} \Rightarrow 1.62 = i \times 4.9 \times \frac{3.9 \times 1000}{122 \times 49} \Rightarrow i = 0.504$. Since $i < 1$, the solute is associated. **Q32.** Increased pressure underwater increases gas solubility in blood. High N_2 solubility causes toxic effects and "bends" during ascent. Helium is less soluble in blood, reducing these risks.

Q33. Henry's law: $P = K_H \cdot \chi$. Solubility (χ) = $\frac{P}{K_H} = \frac{760}{1.25 \times 10^6} = 6.08 \times 10^{-4}$. **Q34.** Moles of water $n_A = \frac{100}{18} = 5.55$. Mole fraction of solvent $\chi_A = \frac{5.55}{1+5.55} = 0.847$.

$P_s = P^\circ \chi_A = 23.80 \times 0.847 = 20.16 \text{ mm Hg}$. **Q35.** Equal moles means $\chi_X = 0.5$ and $\chi_Y = 0.5$. Total Pressure

$P = P_X^\circ \chi_X + P_Y^\circ \chi_Y = 120(0.5) + 160(0.5) = 60 + 80 = 140 \text{ mm Hg}$. **Q36.**

$MgSO_4 \rightarrow Mg^{2+} + SO_4^{2-}$, so $i = 2$. $m = \frac{4 \times 1000}{120 \times 100} = 0.333 \text{ m}$.

$\Delta T_b = i K_b m = 2 \times 0.52 \times 0.333 = 0.346 \text{ K}$. $T_b = 373.15 + 0.346 = 373.496 \text{ K}$. **Q37.**

Molality $m = \frac{19.5 \times 1000}{78 \times 500} = 0.5 \text{ m}$. $\Delta T_f = i K_f m \Rightarrow 1 = i \times 1.86 \times 0.5 \Rightarrow i = 1.075$. For weak acid, $\alpha = \frac{i-1}{n-1}$. Since $n = 2$, $\alpha = \frac{1.075-1}{2-1} = 0.075$ (or 7.5%). **Q38.** Isotonic solutions:

$C_1 = C_2$. $\frac{W_{urea}}{M_{urea}} = \frac{W_{glucose}}{M_{glucose}} \Rightarrow \frac{15}{60} = \frac{W_{glucose}}{180} \Rightarrow W_{glucose} = 45 \text{ g/L}$. **Q39.**

$\Delta T_f(\text{sucrose}) = 273.15 - 269.15 = 4 \text{ K}$. Molality of sucrose = $\frac{10 \times 1000}{342 \times 90}$.

$4 = K_f \times \frac{10000}{342 \times 90} \Rightarrow K_f \approx 12.31$. For 10% glucose: $m = \frac{10 \times 1000}{180 \times 90}$.

$\Delta T_f = 12.31 \times \frac{10000}{180 \times 90} \approx 7.6 \text{ K}$. FP = $273.15 - 7.6 = 265.55 \text{ K}$. **Q40.**

$\Delta T_f = K_f \cdot \frac{W_B \times 1000}{M_B \times W_A} \Rightarrow 0.383 = 3.83 \times \frac{2.56 \times 1000}{M_B \times 100} \Rightarrow M_B = 256 \text{ g mol}^{-1}$.

$32 \times x = 256 \Rightarrow x = 8$. Formula is S_8 . **Q41.** Complete dissociation of $NaCl \rightarrow i = 2$.

$\Delta T_f = i K_f \cdot \frac{W_B \times 1000}{M_B \times W_A} \Rightarrow 2 = 2 \times 1.86 \times \frac{W_B \times 1000}{58.5 \times 37.2} \Rightarrow W_B \approx 1.17 \text{ g}$. **Q42.** Minimum-boiling

azeotrope. E.g., Ethanol + water. Formed when A-B interactions are weaker than A-A and B-B interactions. $\Delta H_{mix} > 0$ (endothermic).

Q43. (A) Properties depending only on the number of solute particles. Molality is preferred because it depends on mass, which doesn't change with temperature, unlike volume in molarity.

(B) $P_A = P_A^\circ \chi_A$. $P_A^\circ - P_A = P_A^\circ(1 - \chi_A) = P_A^\circ \chi_B$. $\frac{\Delta P}{P_A^\circ} = \chi_B = \frac{n_B}{n_A + n_B} \approx \frac{W_B M_A}{M_B W_A}$. **Q44.** (A)

For Na_2SO_4 , $i = 3$.

$\Pi_1 = \Pi_2 \Rightarrow i_1 C_1 = i_2 C_2 \Rightarrow 3 \times \frac{5}{322} = 1 \times \frac{2}{M_X} \Rightarrow M_X = \frac{2 \times 322}{15} \approx 42.93 \text{ g mol}^{-1}$. (B)

Solute lowers solvent's VP. Higher temp is required for VP to equal atmospheric pressure, raising the BP (graph showing pure solvent vs solution crossing 1 atm line). **Q45.** (A) Applying pressure $> \Pi$ to force solvent from solution to pure solvent. Used in desalination. (B) $\Pi = C \cdot R \cdot T$. (C)

$\Pi = \frac{W_B \cdot R \cdot T}{M_B \cdot V} \Rightarrow 2 \times 10^{-3} = \frac{1.0 \times 0.0821 \times 300}{M_B \times 0.200} \Rightarrow M_B = 61,575 \text{ g mol}^{-1}$. **Q46.** (A) Maximum-boiling azeotrope (negative deviation) because strong H-bonds form between acetone and chloroform. (B) $K_3[Fe(CN)_6] \rightarrow 3K^+ + [Fe(CN)_6]^{3-} \Rightarrow n = 4$.

$i = 1 + (n - 1)\alpha = 1 + 3(0.852) = 3.556$. (C)

$\Delta T_b = i K_b m = 3.556 \times 0.52 \times 1 = 1.849 \text{ K}$. BP = $373.15 + 1.849 = 374.999 \text{ K}$ (or 101.85°C). **Q47.** (A) $\Delta T_f = i \cdot K_f \cdot m$. (B) 1 M $MgCl_2$ has higher BP because it yields 3 ions ($i = 3$) compared to glucose ($i = 1$). (C) Molality $m = \frac{5 \times 1000}{122 \times 35} = 1.17 \text{ m}$.

$\Delta T_f = i K_f m \Rightarrow 2.94 = i \times 4.9 \times 1.17 \Rightarrow i = 0.512$. For dimer: $2A \rightarrow A_2$, $\alpha = \frac{1-i}{1-1/n} = \frac{1-0.512}{1-0.5} = \frac{0.488}{0.5} = 0.976$. Association = 97.6%. **Q48.** (A) Partial pressure of gas is proportional to its mole fraction. Applications: CO_2 in soda, Scuba diving tanks. (B)

$P = K_H \cdot \chi \Rightarrow 6 \text{ bar} = 40,000 \text{ bar} \times \chi \Rightarrow \chi = \frac{6}{40000} = 1.5 \times 10^{-4}$. **Q49.** (A) Solution obeying Raoult's law over entire concentration range. $\Delta H_{mix} = 0$, $\Delta V_{mix} = 0$. (B)

$P_T = P_A^\circ \chi_A + P_B^\circ(1 - \chi_A) \Rightarrow 600 = 450\chi_A + 700(1 - \chi_A) \Rightarrow 250\chi_A = 100 \Rightarrow \chi_A = 0.4$, $\chi_B = 0.6$. Vapour phase: $y_A = \frac{P_A}{P_T} = \frac{450 \times 0.4}{600} = 0.3$, $y_B = 0.7$. **Q50.** (A) A-B interactions are stronger than A-A/B-B interactions. E.g., Phenol+Aniline. $\Delta H_{mix} < 0$, $\Delta V_{mix} < 0$. (B)

$CaCl_2 \rightarrow Ca^{2+} + 2Cl^-$ ($i = 3$). Molality $m = \frac{10 \times 1000}{111 \times 200} = 0.45 \text{ m}$.

$\Delta T_b = 3 \times 0.512 \times 0.45 = 0.69 \text{ K}$.